

Q1. DIRECTIONS *for question 1:* Select the correct alternative from the given choices.

In an isosceles triangle, sides AB and AC each measure 63 cm. If D is a point on side BC, such that AD measures 61 cm, find the measure of the product $BD \times DC$ (in sq.cm).

- ☐ a) 244
- ☐ b) 252
- ☐ c) 248
- ☐ d) Cannot be determined

Q2. DIRECTIONS *for question 2:* Type in your answer in the input box provided below the question.

How many pairs of integers exist for which the difference between their sum and product is 42?

Q3. DIRECTIONS *for questions 3 and 4:* Select the correct alternative from the given choices.

A metal sheet in the shape of an equilateral triangle was perfectly cut into a number of smaller equilateral triangles, not necessarily all of the same size. From each of these smaller triangles, a circle of the maximum possible area was cut out. If the thickness of the sheet is uniform, then the total weight of all the circles cut out forms approximately what percentage of the weight of the original triangle?

- ☐ a) **66.67%**
- ☐ b) **68.37%**
- ☐ c) **60.43%**
- ☐ d) **Depends on the number and dimensions of the triangles cut.**

Q4. DIRECTIONS *for questions 3 and 4:* Select the correct alternative from the given choices.

Three pipes, A, B and C, can fill a tank in 40, 60 and 100 minutes respectively. If tap A is opened first, tap B after 10 minutes and tap C after another 5 minutes, in how many more minutes will the tank be filled?

☐ a) 6

☐ b) $15\frac{15}{31}$

☐ c) $10\frac{3}{16}$

☐ d) $10\frac{15}{31}$

Q5. DIRECTIONS *for question 5:* Type in your answer in the input box provided below the question.

Find the value of $\log_{\sqrt{2}} 2^2 + \log_{\sqrt[3]{3}} 3^3 + \dots + \log_{\sqrt[50]{50}} 50^{50}$.

Q6. DIRECTIONS *for question 6:* Select the correct alternative from the given choices.

A cloth merchant, raised the price of cloth in June by $p\%$ over the prices in the previous month. The next month, he announced a discount of $p\%$ over the prices of the previous month. The price of a piece of cloth thus goes down by ₹16, when compared to its price in May. He repeats the same procedure, i.e., in August, he again increases the price of cloth by $p\%$ over the price in July, and in September, he again decreases it by $p\%$ over the previous month. If the price of the same piece of cloth in September is ₹368.64, what was the price of that piece of cloth in the month of May?

☐ a) ₹400

☐ b) ₹360

☐ c) ₹384

☐ d) ₹420

Q7. DIRECTIONS *for question 7:* Type in your answer in the input box provided below the question.

If $N = 364n^5$, where n is a positive integer, has 182 factors, how many possible values can N assume?

Q8. DIRECTIONS *for question 8:* Select the correct alternative from the given choices.

Find the sum of the first 15 terms of the series $1 \times 3 + 2 \times 4 + 3 \times 5 + 4 \times 6 + \dots$

☐ a) 1480

☐ b) 1360

☐ c) 1240

☐ d) 1560

Q9. DIRECTIONS *for question 9:* Type in your answer in the input box provided below the question.

When 952 divides a number, the remainder left is 124. Find the remainder left if 68 divides the same number.

Q10. DIRECTIONS *for questions 10 to 14:* Select the correct alternative from the given choices.

If $x^2 + y^2 + z^2 = 4$, where x, y, z are real numbers, find the range of $(xy + yz + zx)$.

☐ a) **$[-2, 1]$**

☐ b) **$[-2, 2]$**

☐ c) **$[-1, 5]$**

☐ d) **$[-2, 4]$**

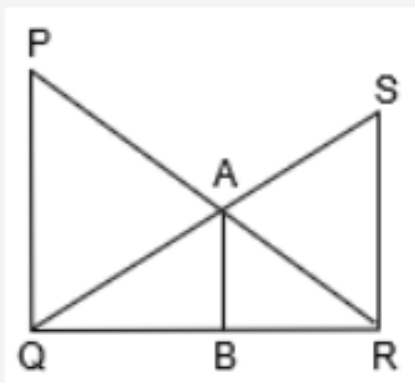
Q11. DIRECTIONS *for questions 10 to 14:* Select the correct alternative from the given choices.

A rope of length 343 m is cut into two pieces, of lengths in the ratio 3 : 4. Each of the pieces obtained is again cut in the ratio 3 : 4. This process is repeated for two more times. If all pieces obtained are kept in the ascending order of their lengths, then what is the length of the seventh piece of rope?

- ☐ a) 15.42 m
- ☐ b) 20.57 m
- ☐ c) 27.47 m
- ☐ d) 18.72 m

Q12. DIRECTIONS for questions 10 to 14: Select the correct alternative from the given choices.

In the given figure, $\angle PQR = \angle SRQ = 90^\circ$ and $AB \perp QR$. If $SR = i$, $PQ = j$ and $QR = k$, find the length of AB .



☐ a) $\frac{i^2 + j^2 + k^2}{i + j + k}$

☐ b) $\frac{ik}{i + k}$

☐ c) $\frac{ij}{i + j}$

☐ d) $\frac{(ij + jk + ki)}{k}$

Q13. DIRECTIONS for questions 10 to 14: Select the correct alternative from the given choices.

For $x > 1$ and $y > 1$, $x \phi y = \frac{2x - y}{x + y}$, when $x > y$, and $x \phi y = xy \phi x$, when $x \leq y$. Find the value of $6 \phi 8$.

☐ a) $\frac{1}{2}$

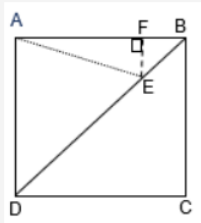
☐ b) $\frac{44}{7}$

☐ c) $\frac{11}{7}$

☐ d) $\frac{5}{3}$

Q14. DIRECTIONS for questions 10 to 14: Select the correct alternative from the given choices.

ABCD is a square of side 4 km. A traveller X starts from D and travels towards B along the diagonal. Simultaneously, another traveller Y starts from A and moves along the dotted line indicated. If at the end of 1 hour both of them meet at E, where $EF = 1$ km ($EF \perp AB$), then X will reach B after a further time of



- ☐ a) $60 - 30\sqrt{3}$ minutes.
- ☐ b) 20 minutes.
- ☐ c) $20\sqrt{3}$ minutes.
- ☐ d) 60 minutes.

Q15. DIRECTIONS *for question 15:* Type in your answer in the input box provided below the question.

If $(2x - 1)(2x - 2)(2x - 3)(2x - 4)(2x - 5) \geq 0$, how many single-digit whole number values can x assume?

Q16. DIRECTIONS *for question 16:* Select the correct alternative from the given choices.

P, Q, R, S and T are five boys who were asked to come, one at a time, and make a presentation. In how many ways is it possible that P makes his presentation before Q, and Q, before R?

☐ a) 10

☐ b) 20

☐ c) 15

☐ d) 30

Q17. DIRECTIONS *for question 17:* Type in your answer in the input box provided below the question.

A certain sum lent at a certain rate of interest per annum, under simple interest, becomes 1.375 times itself in three years. After how many years will the total interest accrued be equal to twice the sum?

Q18. DIRECTIONS for questions 18 to 20: Select the correct alternative from the given choices.

What is the least distance (in units) from the point $(4, 0)$ to the curve $x^2 + y^2 - 6y - 12x + 28 = 0$?

- ☐ a) $\sqrt{13}$
- ☐ b) $\sqrt{17} - \sqrt{13}$
- ☐ c) $\sqrt{17} + \sqrt{13}$
- ☐ d) $2\sqrt{17} - 3\sqrt{13}$

Q19. DIRECTIONS *for questions 18 to 20:* Select the correct alternative from the given choices.

If x and y are positive integers, find the value of $3x - 2y$, given $x^2 y^3 = 256$ and $x > y$.

- ☐ a) 46
- ☐ b) 48
- ☐ c) 50
- ☐ d) **Cannot be determined**

Q20. DIRECTIONS *for questions 18 to 20:* Select the correct alternative from the given choices.

A vendor bought a certain number of kites. Three-fifths of them were white and the rest were yellow. He sold three-sevenths of the kites, which included one-sixth of the yellow kites. What fraction of the unsold kites were white?

☐ a) $\frac{2}{3}$

☐ b) $\frac{3}{5}$

☐ c) $\frac{4}{7}$

☐ d) $\frac{5}{12}$

Q21. DIRECTIONS *for question 21:* Type in your answer in the input box provided below the question.

If $(4 + x + x^2)^8 = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + \dots + a_{16}x^{16}$, find the value of $a_1 + a_3 + a_5 + \dots + a_{15}$.

Q22. DIRECTIONS *for questions 22 and 23:* Select the correct alternative from the given choices.

Nirvaan has a certain number of chocolates with him. He would be left with 7 chocolates, if he distributes them equally among 11 students. If there were 2 more students, he would be left with 5 more chocolates. If Nirvaan has less than 500 chocolates, which of the following cannot be the number of chocolates left with him, if he were to distribute the chocolates among 15 students?

☐ a) 0

☐ b) 6

☐ c) 12

☐ d) 14

Q23. DIRECTIONS *for questions 22 and 23:* Select the correct alternative from the given choices.

The ratio of the present ages of Ajay and Vijay is 7 : 3. If thirteen years ago, the ratio of their ages was 9 : 2, what would be Vijay's age 16 years from now?

☐ a) 37

☐ b) 36

☐ c) 28

☐ d) 43

Q24. DIRECTIONS *for question 24:* Type in your answer in the input box provided below the question.

N marbles can be distributed equally among P children, where $N > P$. What is the number of values that N can assume such that $P > 1$ and $2 < N + P < 100$?